

SAGAR SHAH

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EDUCATION

B.S. in Computer Science and Mechanical Engineering @ University of Virginia

August 2026 - June 2028/29

Fullerton College, Cypress College, Santiago Canyon College, Santa Ana College

June 2020 - 2025

6 Degrees | Student of Distinction in Academic Achievement

- Associates of Science: Computer Science, Math, Physics, Engineering
- Associates of Arts: General Education with an Emphasis on Math and Science
- Associates of Arts: General Education with an Emphasis in Art and Human Expression

Buena Vista Virtual High School | Top 10% of Class of 30

August 2021 - 2025

Notable Coursework at Community College:

- Computer Science: Data Structures and Algorithms in C++, C For Math and Science
- Math: Discrete Mathematics, Graph Theory, Multivariable Calculus, Differential Equations and Linear Algebra
- Physics (Calculus Based): General Physics 1, General Physics 2, General Physics 3
- Engineering: Circuits, AutoCAD, Intro to Engineering

Programming Languages: Python (PyTorch, Numpy, SciKit, Matplotlib), C++, C, Julia, Java, LaTeX, Linux, Swift

Frameworks and Tools: React, Node.js, TensorFlow, Docker, Git, ROS2

Engineering Programs: Fusion 360, OnShape, AutoCAD, Arduino, Raspberry Pi, Ansys

PUBLICATIONS

- Strategic Evasion Using Multi-Hypothesis Control Barrier Functions – AIAA 2026 Aviation – **Sagar Shah**, Isaac Weintraub, Alexander Von Moll – <https://arc.aiaa.org/doi/10.2514/6.2026-4495>

EXPERIENCE

Undergraduate Researcher | CRAL Lab @ University of Virginia Link Lab

September 2026 - Present

- Developing novel control barrier functions for agile bipedal and quadruped robots under Dr. Rohan Chandra in the CRAL.

United States Air Force Research Lab | Air Warfare and 711 Human Performance Wing Directorates **Summer 2024 - Present**

Student Researcher, Wright Patterson AFB, OH

- Leading development of GNC strategies for the Air Warfare Directorate using advanced control theory algorithms written in Julia (First author publication; Gave podium talk at AIAA Aviation 2026)
- Programming a Boston Dynamics Spot Robot Dog in the 711 Human Performance Wing Directorate (CAMS Lab)
 - Autonomously cut a ribbon using computer vision for the lab's inauguration ceremony (<https://tinyurl.com/57z42b7p>)
 - Implemented SLAM and ROS2 Humble (non-native to Spot) for multi-domain accessibility with UR Robot Arms
- Created a GUI compatible with ARES OS for a Soft Materials lab in the Materials Directorate (code is published and in use)

BioDesign Consultant (Paid Staff) | Mussallem Center for Biodesign @ Stanford University

September 2025 - Present

- Led development of pediatric wearable devices for patients in the NICU/PICU post-cardiovascular surgery at Stanford Hospital
 - Worked with BLE-enabled microcontroller to pair with EKGs and device for continuous monitoring; Written in C and C++
- Created tutorials aimed to equip Biodesign attendees with the practical skills and confidence to conceptualize, prototype, and evaluate early-stage digital health solutions using modern AI techniques. Taught how to implement ML in Google Colab.

Software Engineering Contracting Intern | BAE Systems United Kingdom

October 2025

- Month long contract to work on integrated navigation systems on next-gen unmanned aerial vehicles (UAVs) – NDAs-signed

Student Researcher | Fuller Lab @ Stanford University School of Engineering

Spring 2023 - Fall 2024

- Developed machine learning pipeline using computer vision for pupil detection utilizing Meta's SAM 2 AI Model using PyTorch
- Worked in team to develop an app that detects ophthalmological disease (deployed to Alcon) using React.js and TypeScript

NASA MINDS Competition – Grand Champions (Team of 10 students)

Spring 2024 - Fall 2024

- Developed RFID based real-time object tracking system (Echo and Eureka Systems) using triangulation and A* algorithms
- Lead Programmer – Wrote system algorithms & GUI of Echo & Eureka systems using C++ and Python
- Presented it to NASA panel, consisting of engineers and astronauts, from HQ and different space centers (KSC, JPL, JSC, etc.)

Applied Engineering Club | President @ Fullerton College

Fall 2023 - Spring 2024

- Led the software and mechanical design of a UAV using ROS stacks and Ansys for the C-UAS UAV competition at CSULA
- Autonomous Drone Racing Competition (A2RL)
 - One of the lead programmers for developing a fully autonomous racing drone for the A2RL competition held in Dubai.

HONORS AND AWARDS

- US Air Force Wright Scholar (Summer 2024, Summer 2025, Summer 2026 – 1 of 20 out of 800 applicants)
- NASA MINDS Grand Champions – 1 of 50 schools to be selected nationally – Project concept deployed to Artemis II Mission
- College Reading and Learning Association (CRLA) 1 Certified – Fullerton College – International English tutor certificate
- 2025 Fullerton College Student of Distinction—1 of 5 out of all 2025 graduates to receive scholarship for academic achievement
- Speech and Debate – Toastmasters International – Silver Medalist (1 rank from Gold) – Spoke in front of 300+ people